

Low-Temperature Induction Reduces Inclusion-Body Formation of SynLig-B Without Changing Total Expression

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ABSTRACT

SynLig-B was expressed for 6 h at 30 C or for 18 h at 16 C after induction with 0.1 mM IPTG. Total SynLig-B expression was similar between conditions, but the soluble fraction increased from 18% at 30 C to 57% at 16 C. Volumetric soluble yield increased from 1.9 to 5.8 mg/L. Lower temperature did not increase the total amount of SynLig-B and did not change the apparent molecular mass. The result reflects improved partitioning into the soluble fraction.

Keywords: *induction temperature, inclusion body, soluble fraction, SynLig-B, OCR benchmark*

INTRODUCTION

Lower induction temperature is commonly used to slow synthesis and reduce aggregation. However, a thicker soluble band can arise either from increased total expression or from redistribution of a fixed amount of target protein.

The present study quantified total, soluble, and pellet-associated SynLig-B under two induction temperatures. The PDF is intentionally formatted as a degraded scan to test OCR and reading-order robustness.

MATERIALS AND METHODS

Expression conditions

A synthetic *E. coli* strain was induced with 0.10 mM IPTG at optical density 0.7. One group was incubated at 30 C for 6 h and the other at 16 C for 18 h. Culture volume and sampling mass were matched.

Fractionation

Cells were disrupted by sonication. Equal total protein from whole lysate, clarified supernatant, and washed pellet was analyzed by SDS-PAGE. SynLig-B was quantified with a purified calibration standard.

Calculated outcomes

Soluble fraction was defined as soluble SynLig-B divided by total SynLig-B. Volumetric soluble yield was calculated per liter of culture.

Condition	IPTG	Time	Temperature
Warm	0.10 mM	6 h	30 C
Cool	0.10 mM	18 h	16 C

Table 1. Induction conditions.

RESULTS

Total expression was similar

Total SynLig-B was 10.4 +/- 0.9 mg/L at 30 C and 10.1 +/- 1.0 mg/L at 16 C. The difference was less than 3%. Apparent molecular mass was 82 kDa in both groups.

Solubility improved at 16 C

The soluble fraction increased from 18 +/- 4% at 30 C to 57 +/- 6% at 16 C. Soluble yield increased from 1.9 +/- 0.4 to 5.8 +/- 0.7 mg/L, while pellet-associated target decreased.

Growth trade-off

Final wet cell weight was 8% lower at 16 C, but this did not offset the increase in soluble product.

Outcome	30 C	16 C
Total SynLig-B (mg/L)	10.4 +/- 0.9	10.1 +/- 1.0
Soluble fraction	18 +/- 4%	57 +/- 6%
Soluble yield (mg/L)	1.9 +/- 0.4	5.8 +/- 0.7

Table 2. Lower temperature changed solubility, not total expression.

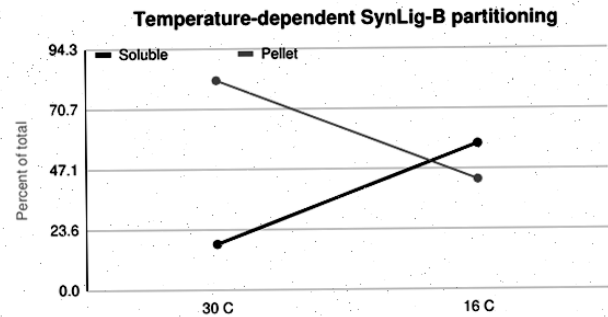


Figure 1. The target redistributed from pellet to soluble fraction at 16 C.

DISCUSSION

Low-temperature induction improved SynLig-B solubility without increasing total expression. The threefold increase in soluble yield is therefore a partitioning effect rather than greater synthesis.

The different induction durations are part of the process comparison and prevent attribution to temperature alone at a mechanistic level.

LIMITATIONS

Temperature and induction time changed together. Folding kinetics, chaperone binding, and mRNA abundance were not measured.

CONCLUSION

Induction at 16 C increased the soluble fraction of SynLig-B from 18% to 57% and soluble yield from 1.9 to 5.8 mg/L, while total expression remained essentially unchanged.

REFERENCES

The following references are fictional and are included only to test reference-section handling.

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